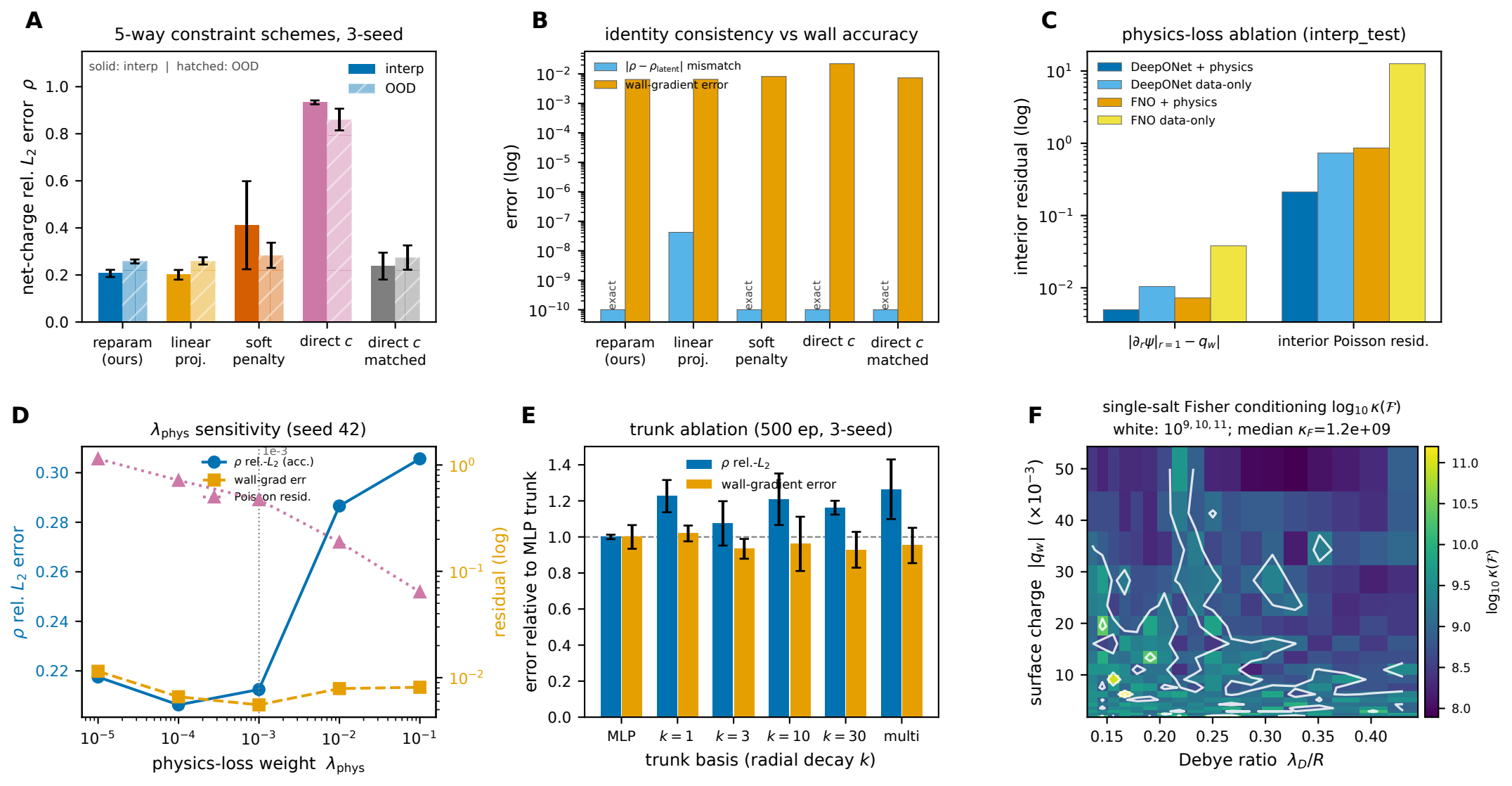
A
5-way constraint schemes, 3-seed
solid: interp | hatched: OOD
interp
OOD
net-charge rel. L_2 error ρ
reparam (ours)
linear proj.
soft penalty
direct c
direct c matched
B
identity consistency vs wall accuracy
$|\rho - \rho_{\text{latent}}|$ mismatch
wall-gradient error
error (log)
exact
C
physics-loss ablation (interp_test)
DeepONet + physics
DeepONet data-only
FNO + physics
FNO data-only
interior residual (log)
$|\partial_r \psi|_{r=1} - q_w|$
interior Poisson resid.
D
λ_{phys} sensitivity (seed 42)
1e-3
ρ rel.-L_2 (acc.)
wall-grad err
Poisson resid.
ρ rel. L_2 error
residual (log)
physics-loss weight λ_{phys}
E
trunk ablation (500 ep, 3-seed)
ρ rel.-L_2
wall-gradient error
error relative to MLP trunk
MLP
$k=1$
$k=3$
$k=10$
$k=30$
multi
trunk basis (radial decay k)
F
single-salt Fisher conditioning $\log_{10}\kappa(\mathcal{F})$
white: $10^{9,10,11}$; median κ_F=1.2e+09
surface charge $|q_w|$ ($\times 10^{-3}$)
Debye ratio λ_D/R
$\log_{10}\kappa(\mathcal{F})$